

# DATA 202 — Quiz 1

## Week 2 — DataFrame Basics & Visual Encodings

Name: \_\_\_\_\_

Date: \_\_\_\_\_

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### The Dataset: Product Sales

`sales` contains 50 simulated products. Each row describes one product's performance: how much it sold, how it was advertised, and how profitable it was.

**Columns:** Product ID, Sales, Units Sold, Profit, Advertising Spend (numerical)  
· Category, Season, Supplier (categorical)

#### First few rows:

| Product ID | Sales | Units Sold | Profit | Advertising Spend | Category    | Season | Supplier   |
|------------|-------|------------|--------|-------------------|-------------|--------|------------|
| P001       | 99268 | 684        | 17337  | 5581              | Electronics | Summer | Supplier A |
| P003       | 71613 | 924        | 3832   | 2788              | Furniture   | Winter | Supplier A |
| P006       | 72993 | 1239       | 10683  | 6070              | Clothing    | Summer | Supplier A |
| P011       | 46589 | 743        | 1109   | 6384              | Furniture   | Fall   | Supplier B |
| P026       | 98755 | 1206       | 18346  | 3195              | Electronics | Spring | Supplier C |

### Question 1 (SLO 02A)

For each command below, identify whether it **Accesses**, **Adds**, or **Deletes** data, and whether it operates on a **Row** or a **Column**.

| # | Command                                                                             | Access / Add / Delete? | Row(s) or Column? |
|---|-------------------------------------------------------------------------------------|------------------------|-------------------|
| 1 | <code>sales["Advertising Spend"]</code>                                             |                        |                   |
| 2 | <code>sales["Discount"] = sales["Sales"] * 0.1</code>                               |                        |                   |
| 3 | <code>sales = pd.concat([sales, pd.DataFrame([new_row])], ignore_index=True)</code> |                        |                   |
| 4 | <code>sales = sales[sales["Category"] != "Furniture"]</code>                        |                        |                   |
| 5 | <code>sales = sales.drop(columns=["Returns"])</code>                                |                        |                   |

### Question 2 (SLO 02B)

Complete the code below to select all products in the **Electronics** category with **Advertising Spend over \$5,000**, sorted by Advertising Spend from highest to lowest.

*Remember: the filter condition still goes inside the square brackets and must reference `sales` again, e.g. `sales[sales["Column"] ...]`.*

```
result = sales[
    _____
]
result = result.sort_values("_____", ascending=_____)
print(len(result))
```

### Question 3 (SLO 02C)

Available visual channels: `x`, `y`, `color`, `size`, `symbol`, `facet_col`.

Since this is a paper quiz, `color` isn't something you can sketch in pencil — the code below uses `symbol` instead, which maps a categorical column to different *marker shapes* (e.g.  $\times$ ,  $\circ$ ,  $\triangle$ ) rather than colors.

For the code below, fill in the table: for each visual channel it uses, name the column mapped to it and say whether that column is Numerical or Categorical.

```
px.scatter(sales, x="Advertising Spend", y="Profit",
           symbol="Category", size="Units Sold",
           title="Advertising Spend vs. Profit")
```

| Visual Channel | Column Used | Data Type (Numerical / Categorical) |
|----------------|-------------|-------------------------------------|
| x-axis         |             |                                     |
| y-axis         |             |                                     |
| symbol         |             |                                     |
| size           |             |                                     |

In the space below, sketch what you expect this chart to look like. Label your axes. You don't need exact data — just show the general pattern (point cloud, marker shapes ( $\times/\circ/\triangle$ ) for groups, varying point sizes).